Assignment 3 — HTML Iframes, Forms & Head Elements

Complete worked solutions for all 20 questions. Each solution is a self-contained, pure HTML file (no CSS/JavaScript), formatted with indentation and explanatory comments as required.

1. YouTube Video Embed (percentage width/height)
2. Google Map Embed
3. Two Stacked Iframes (Intro + Hobby)
4. Frameset Layout (Top Nav / Bottom Message)
5. Complete Head Section + Multilingual Paragraph
6. User Registration Form
7. Login Form with CSRF Token
8. Feedback Form
9. File Upload Form (.jpg / .png only)
10. Contact Form
11. Student Information Form (Dropdown)
12. HTML Entities Paragraph
13. Three Language Page (English, Hindi, French)
14. External Website Embed
15. Payment Form
16. URI Encoding Demonstration
17. Form + Result Page Loaded in Iframe
18. Long and Short Quotations
19. Form Security Attributes (autocomplete off)
20. Complete Head Elements Demo

Question 1. YouTube Video Embed (percentage width/height)

File: q1_youtube.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q1 - YouTube Video Embed</title>
</head>
<body>
    <h1>My Favorite YouTube Video</h1>

    <!-- The iframe uses percentage values for width and height so it -->
    <!-- resizes automatically with the browser window -->
    <iframe
        src="https://www.youtube.com/embed/dQw4w9WgXcQ"
        width="80%"
        height="60%"
        title="YouTube video player"
        allowfullscreen>
    </iframe>
</body>
</html>
```

Question 2. Google Map Embed

File: q2_googlemap.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q2 - Google Map Embed</title>
</head>
<body>
    <h1>Location of My School</h1>

    <!-- loading="lazy" tells the browser to delay loading the iframe's -->
    <!-- content (the map) until it is about to appear in the viewport. -->
    <!-- This improves page load speed and saves bandwidth. -->
    <iframe
        src="https://www.google.com/maps/embed?pb=!1m18!1m12!1m3!1d224345.7!2d77.6737!3d27.4924!2m3!1f0!2f0!3f0!3m2!1i1024!2i768!4f13.1"
        title="Map showing the location of my school"
        width="90%"
        height="70%"
        loading="lazy">
```

```
    </iframe>
</body>
</html>
```

Question 3. Two Stacked Iframes (Intro + Hobby)

File: q3_two_iframes.html

> Assumes two supporting files, intro.html and hobby.html, exist in the same folder and contain a short paragraph each.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q3 - Two Iframes Stacked</title>
</head>
<body>
    <h1>About Me</h1>

    <!-- First iframe: loads the introduction page -->
    <!-- (intro.html must exist in the same folder) -->
    <h2>Introduction</h2>
    <iframe src="intro.html" width="600" height="250" title="My Introduction">
    </iframe>

    <!-- Second iframe: loads the favorite hobby page, placed below the first -->
    <!-- (hobby.html must exist in the same folder) -->
    <h2>My Hobby</h2>
    <iframe src="hobby.html" width="600" height="250" title="My Favorite Hobby">
    </iframe>
</body>
</html>
```

Question 4. Frameset Layout (Top Nav / Bottom Message)

File: q4_frameset.html

> Two supporting files are required: nav.html (with Home, About, Contact links) and welcome.html (a page containing the text "Welcome to My Website"). Their code is shown below.

```
<!DOCTYPE html>
<html lang="en">
<head>
    <title>Q4 - Frameset Layout</title>
</head>
<!-- The <frameset> tag replaces <body> and splits the window into rows -->
<frameset rows="20%,80%">
    <!-- Top frame: navigation links -->
    <frame src="nav.html" name="topFrame" title="Navigation">
    <!-- Bottom frame: welcome message -->
    <frame src="welcome.html" name="bottomFrame" title="Welcome Message">

    <noframes>
        <body>Your browser does not support frames.</body>
    </noframes>
</frameset>
</html>
```

4A. Supporting file: nav.html

File: nav.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Navigation</title>
</head>
<body>
    <!-- Simple navigation links shown in the top frame -->
    <a href="home.html" target="bottomFrame">Home</a> |
    <a href="about.html" target="bottomFrame">About</a> |
    <a href="contact.html" target="bottomFrame">Contact</a>
</body>
</html>
```

4B. Supporting file: welcome.html

File: welcome.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Welcome</title>
</head>
<body>
    <h2>Welcome to My Website</h2>
</body>
</html>
```

Question 5. Complete Head Section + Multilingual Paragraph

File: q5_head_utf8.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <!-- Ensures special/multilingual characters render correctly -->
    <meta charset="UTF-8">

    <!-- Identifies the author of the page -->
    <meta name="author" content="Student Name">

    <!-- Makes the page responsive on mobile devices -->
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>Q5 - UTF-8 Encoding Test</title>
</head>
<body>
    <h1>UTF-8 Encoding Test</h1>

    <!-- Paragraph containing three languages to verify UTF-8 encoding -->
    <p>
        English: Welcome to my website.<br>
        Hindi: &#2350;&#2375;&#2352;&#2368; &#2357;&#2375;&#2348;&#2360;&#2366;&#2311;&#2335; &#2350;&#2375;&#2306; &#2310;&#2346;&#2325;&#2366;
&#2360;&#2381;&#2357;&#2366;&#2327;&#2340; &#2361;&#2376;&#2404;<br>
        Japanese: &#12371;&#12398;&#12454;&#12455;&#12502;&#12469;&#12452;&#12488;&#12408;&#12424;&#12358;&#12371;&#12381;
    </p>
</body>
</html>
```

Question 6. User Registration Form

File: q6_registration.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q6 - User Registration Form</title>
</head>
<body>
    <h1>User Registration</h1>

    <form method="post" action="#">
        <!-- All fields use the required attribute -->
        <label for="fullname">Full Name:</label><br>
        <input type="text" id="fullname" name="fullname" required><br><br>

        <label for="email">Email Address:</label><br>
        <input type="email" id="email" name="email" required><br><br>

        <label for="password">Password:</label><br>
        <input type="password" id="password" name="password" required><br><br>

        <label for="confirmpassword">Confirm Password:</label><br>
        <input type="password" id="confirmpassword" name="confirmpassword" required><br><br>

        <input type="submit" value="Register">
    </form>
</body>
</html>
```

Question 7. Login Form with CSRF Token

File: q7_login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q7 - Login Form</title>
```

```
</head>
<body>
    <h1>Login</h1>

    <form method="post" action="#">
        <label for="username">Username:</label><br>
        <input type="text" id="username" name="username" required><br><br>

        <label for="password">Password:</label><br>
        <input type="password" id="password" name="password" required><br><br>

        <!-- A CSRF (Cross-Site Request Forgery) token is a hidden, unique   -->
        <!-- value generated by the server for each session/form. It is     -->
        <!-- sent back with the form submission so the server can verify    -->
        <!-- that the request genuinely came from its own form and not      -->
        <!-- from a malicious third-party site pretending to be the user.   -->
        <input type="hidden" name="csrf_token" value="a1b2c3d4e5f6dummy">

        <input type="submit" value="Login">
    </form>
</body>
</html>
```

Question 8. Feedback Form

File: q8_feedback.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q8 - Feedback Form</title>
</head>
<body>
    <h1>Feedback Form</h1>

    <form method="post" action="#">
        <label for="feedback">Your Feedback:</label><br>
        <textarea id="feedback" name="feedback" rows="5" cols="40"></textarea><br><br>

        <p>Rating:</p>
        <input type="radio" id="good" name="rating" value="Good">
        <label for="good">Good</label><br>

        <input type="radio" id="average" name="rating" value="Average">
        <label for="average">Average</label><br>

        <input type="radio" id="poor" name="rating" value="Poor">
        <label for="poor">Poor</label><br><br>

        <input type="submit" value="Submit Feedback">
    </form>
</body>
</html>
```

Question 9. File Upload Form (.jpg / .png only)

File: q9_fileupload.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q9 - File Upload Form</title>
</head>
<body>
    <h1>Upload Your Photo</h1>

    <form method="post" action="#" enctype="multipart/form-data">
        <label for="photo">Choose an image:</label><br>
        <!-- accept attribute restricts the file picker to .jpg and .png files -->
        <input type="file" id="photo" name="photo" accept=".jpg,.png" required><br><br>
        <input type="submit" value="Upload">
    </form>

    <p>Note: Only .jpg and .png image files are allowed for upload.</p>
</body>
</html>
```

Question 10. Contact Form

File: q10_contact.html

```
<!DOCTYPE html>
```

```
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q10 - Contact Form</title>
</head>
<body>
    <h1>Contact Us</h1>

    <form method="post" action="#">
        <label for="name">Full Name:</label><br>
        <input type="text" id="name" name="name" placeholder="Enter your full name" required><br><br>

        <label for="email">Email:</label><br>
        <input type="email" id="email" name="email" placeholder="Enter your email address" required><br><br>

        <label for="subject">Subject:</label><br>
        <input type="text" id="subject" name="subject" placeholder="Enter the subject" required><br><br>

        <label for="message">Message:</label><br>
        <textarea id="message" name="message" rows="5" cols="40" placeholder="Type your message here" required></textarea><br><br>

        <input type="submit" value="Send Message">
    </form>
</body>
</html>
```

Question 11. Student Information Form (Dropdown)

File: q11_student_info.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q11 - Student Information Form</title>
</head>
<body>
    <h1>Student Information</h1>

    <form method="post" action="#">
        <label for="sname">Name:</label><br>
        <input type="text" id="sname" name="sname" required><br><br>

        <label for="roll">Roll Number:</label><br>
        <input type="text" id="roll" name="roll" required><br><br>

        <label for="course">Course:</label><br>
        <select id="course" name="course" required>
            <!-- Default disabled option acts as a placeholder -->
            <option value="" disabled selected>Select Course</option>
            <option value="bca">BCA</option>
            <option value="bsc">B.Sc Computer Science</option>
            <option value="btech">B.Tech</option>
            <option value="mca">MCA</option>
        </select><br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

Question 12. HTML Entities Paragraph

File: q12_entities.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q12 - HTML Entities</title>
</head>
<body>
    <h1>HTML Entities Demo</h1>

    <p>
        Less than: &lt;    Greater than: &gt;   
        Ampersand: &amp;    Double quote: &quot;   
        Single quote: &apos;
    </p>

    <p>
        Copyright: &copy;    Trademark: &trade;   
        Registered: &reg;
    </p>
</body>
</html>
```

Question 13. Three Language Page (English, Hindi, French)

File: q13_multilang.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <!-- Ensures Hindi and French characters display correctly -->
    <meta charset="UTF-8">
    <title>Q13 - Multilingual Page</title>
</head>
<body>
    <h1>Greetings in Three Languages</h1>

    <p>English: Hello, welcome to my website!</p>
    <p>Hindi: &#2344;&#2350;&#2360;&#2381;&#2340;&#2375;, &#2350;&#2375;&#2352;&#2368; &#2357;&#2375;&#2348;&#2360;&#2366;&#2311;&#2335;
&#2346;&#2352; &#2310;&#2346;&#2325;&#2366; &#2360;&#2381;&#2357;&#2366;&#2327;&#2340; &#2361;&#2376;!</p>
    <p>French: Bonjour, bienvenue sur mon site web !</p>
</body>
</html>
```

Question 14. External Website Embed

File: q14_external_embed.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q14 - Embedding an External Website</title>
</head>
<body>
    <h1>Embedding Wikipedia</h1>

    <iframe
        src="https://www.wikipedia.org"
        title="Wikipedia Website"
        width="90%"
        height="500">
    </iframe>

    <p>Some websites may not allow embedding for security reasons.</p>
</body>
</html>
```

Question 15. Payment Form

File: q15_payment.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q15 - Payment Form</title>
</head>
<body>
    <h1>Payment Details</h1>

    <form method="post" action="#">
        <label for="cardname">Cardholder Name:</label><br>
        <input type="text" id="cardname" name="cardname" required><br><br>

        <label for="cardnumber">Card Number:</label><br>
        <input type="text" id="cardnumber" name="cardnumber" maxlength="16" required><br><br>

        <label for="expiry">Expiry Date:</label><br>
        <input type="month" id="expiry" name="expiry" required><br><br>

        <label for="cvv">CVV:</label><br>
        <input type="password" id="cvv" name="cvv" maxlength="3" required><br><br>

        <input type="submit" value="Pay Now">
    </form>
</body>
</html>
```

Question 16. URI Encoding Demonstration

File: q16_uri_encoding.html

```
<!DOCTYPE html>
<html lang="en">
```

```
<head>
    <meta charset="UTF-8">
    <title>Q16 - URI Encoding</title>
</head>
<body>
    <h1>URI Encoding Example</h1>

    <!-- The space characters in the query string are encoded as %20 -->
    <!-- so the URL is valid, since raw spaces are not allowed in a URI -->
    <a href="https://example.com/search?query=HTML%20&%20CSS">
        Search for HTML &amp; CSS
    </a>

    <p>Original link: https://example.com/search?query=HTML &amp; CSS</p>
    <p>Encoded link: https://example.com/search?query=HTML%20&amp;%20CSS</p>
</body>
</html>
```

Question 17. Form + Result Page Loaded in Iframe

File: q17_form.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q17 - form.html</title>
</head>
<body>
    <h1>Submit the Form</h1>

    <!-- The form's target is set to the iframe's name, so the response -->
    <!-- (result.html) loads inside the iframe instead of the whole page -->
    <form method="get" action="result.html" target="resultFrame">
        <label for="text">Enter Text:</label>
        <input type="text" id="text" name="text" required>
        <input type="submit" value="Submit">
    </form>

    <h2>Result:</h2>
    <iframe name="resultFrame" src="" width="100%" height="150" title="Form Result">
    </iframe>
</body>
</html>
```

17A. Supporting file: result.html

File: result.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q17 - result.html</title>
</head>
<body>
    <p>Form successfully submitted.</p>
</body>
</html>
```

Question 18. Long and Short Quotations

File: q18_quotations.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q18 - Quotations</title>
</head>
<body>
    <h1>Quotation Examples</h1>

    <!-- Long quotation -->
    <blockquote>
        The only way to do great work is to love what you do &amp; never stop learning.
    </blockquote>

    <!-- Short quotation using <q>, with an HTML entity inside -->
    <p>
        As the saying goes, <q>knowledge &amp; wisdom go hand in hand</q>, and it
        reminds us to keep growing every day.
    </p>
</body>
```

```
</html>
```

Question 19. Form Security Attributes (autocomplete off)

File: q19_security.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Q19 - Form Security Attributes</title>
</head>
<body>
    <h1>Secure Login Form</h1>

    <!-- autocomplete="off" prevents the browser from storing and -->
    <!-- suggesting previously entered values for this form -->
    <form method="post" action="#" autocomplete="off">
        <label for="username">Username:</label><br>
        <input type="text" id="username" name="username" required><br><br>

        <label for="password">Password:</label><br>
        <input type="password" id="password" name="password" required><br><br>

        <input type="submit" value="Login">
    </form>

    <p>
        Turning off autocomplete can improve security because it stops
        sensitive information (like passwords) from being saved in the
        browser and suggested later, which reduces the risk of that
        information being exposed on a shared or public computer.
    </p>
</body>
</html>
```

Question 20. Complete Head Elements Demo

File: q20_head_elements.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <!-- Sets the character encoding for the document -->
    <meta charset="UTF-8">

    <!-- Provides a short description of the page for search engines -->
    <meta name="description" content="A demo page showing all important head elements">

    <!-- Identifies the author of the page -->
    <meta name="author" content="Student Name">

    <!-- Sets the base URL for all relative links/resources on the page -->
    <base href="https://example.com/">

    <!-- Links an external resource; here, the site's favicon -->
    <link rel="icon" href="favicon.ico" type="image/x-icon">

    <!-- Title shown on the browser tab -->
    <title>Q20 - Head Elements Demo</title>

    <!-- Empty style tag reserved for internal CSS (left empty per instructions) -->
    <style></style>

    <!-- Empty script tag reserved for JavaScript (left empty per instructions) -->
    <script></script>
</head>
<body>
    <h1>Complete Head Section Demo</h1>
    <p>This page demonstrates all the important elements inside the &lt;head&gt; section.</p>
</body>
</html>
```